INCH-POUND

GGG-V-790E
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SUPERSEDING
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August 27, 1985

FEDERAL SPECIFICATION

VULCANIZERS, HOT PATCH AND PLATE, INNER TUBE

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers an electrically heated vulcanizer plate for applying tube patches and rubber-covered-base valve stems to inner tubes.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Standards

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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(Single copies of this specification, and other Federal Specifications and commercial item description is required by activities outside the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	-	Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-V-45554	-	Vulcanizing Equipment, Including Related Items, Tire and Tube Rebuild and Repair, Preparation For Delivery of

Military Standards

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation:

Occupational Health and Safety Administration (OSHA) 29 CFR 1910.211-222
General Industrial Safety and Health Standards

(Copies of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging
ASTM E380-84 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02669.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more vulcanizers for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The vulcanizers shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The vulcanizer shall be designed to accomplish tube repairs without damage to the tubes. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the vulcanizer or any vulcanizer part or component.

3.2.1 Measurement system. The US Customary system of Units (US) or the International System of Units (SI) may be used in the design and construction of the vulcanizers. In this specification, all measurements, dimensions,

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sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the vulcanizer or with the operator's performance.

3.2.2.1 Occupational safety standards. The product shall comply with OSHA standards set forth in 29 CFR 1910.211-222 that are applicable to vulcanizers.

3.2.3 Interchangeability. Parts shall be designed and manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. Parts used in fabrication of the vulcanizers shall be new and free of defects. Part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Screws, pins, bolts, and similar parts shall be installed with means to prevent loss of tightness and adjustment. Parts subject to adjustment or removal shall not be swaged, peened, staked, or otherwise deformed.

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Construction of the vulcanizers shall be rugged and sturdy, and indicative of equipment produced by manufacturers normally involved in producing this type of equipment.

3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.4.4 Protective finishes. All exposed metal (see 6.5.1) and working surfaces, except those made of corrosion-resistant materials, shall be coated or covered with a protective finish that provides protection against corrosion.

3.5 Performance and product characteristics.

3.5.1 Vulcanizer. The vulcanizer shall be electrically-heated, bench or floor mounted, which incorporate facilities for holding and curing tube patches and rubber-covered-base valve stems onto inner tubes of sizes up to and including, 14.00-24. The vulcanizer shall be suitable for repairing one inner tube at one time. The curing head on the vulcanizer shall be a minimum of 12 inches long by 4 inches wide. The time required for bringing the electrically-heated vulcanizers to vulcanizing temperature prior to performing the repair operations shall not exceed 30 minutes.

3.6 Components.

3.6.1 Clamping unit. The vulcanizer shall be provided with a single hand-operated, adjustable screw or quick-acting clamp. The clamp shall hold the tube and patch in position without loosening during the vulcanizing process.

3.6.2 Plates. The vulcanizer shall be equipped with plates, pads and spiders, for tube patch and valve stem repairs on inner tubes, suitable for the vulcanizer.

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3.6.3 Electrical equipment. Electrical equipment used on the vulcanizer shall conform to NFPA 70.

3.6.3.1 Heating. The vulcanizer shall be electrically heated. The heating elements shall be of the resistance type. The heating elements shall be insulated to prevent current leakage and shall distribute heat evenly throughout the vulcanizing plates. The heating elements shall be operated from 115 volts, 60 hertz, single-phase current. The time required to bring the vulcanizer to vulcanizing temperature shall not exceed 30 minutes.

3.6.3.2 Automatic control. Electrically-heated vulcanizers shall be equipped with an automatic temperature control which shall be adjustable to any setting from +287 degrees to +312 degrees Fahrenheit (F). The automatic control shall maintain the set temperature within ±5 degrees F at any setting.

3.6.3.3 Wiring. Wiring used either within, or adjacent to, heat zones of the electrically-heated vulcanizers shall be protected from the heat by being suitably insulated. Exposed wiring shall be of multiple-conductor type encased in a protective sheath. Wiring shall be of a size that shall carry the maximum sustained current requirements of the vulcanizer without damage.

3.6.4 Temperature indicator. The vulcanizer shall be furnished with a temperature indicator to indicate when the plates are at vulcanizing temperature.

3.6.5 Accessory equipment. A roughener rasp or a hand-operated wire buffer shall be furnished with each vulcanizer. The tool shall be attached with a chain, or other suitable means, to the vulcanizer to prevent loss.

3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent, (i.e., epoxy enamel), shall be applied.

3.9 Product identification. For military activities, the vulcanizers shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number. For

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civil agencies, the vulcanizers shall be marked in a plain and permanent manner with the manufacturer's name or trademark in such a way that the source of manufacture can be readily identified.

3.10 Workmanship. Standards of workmanship shall assure that vulcanizers shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see

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6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Temperature control and accuracy test (see 4.6.2).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed ass listed with all samples selected randomly.

Inspection or Test	Lot or Batch Size	Sample Size
Product examination (4.5)	1 - 150	100 percent
Temperature control and accuracy test (4.6.2)	1 - 150	13 percent
Packaging inspection (4.7)	1 - 150	5 percent

The lot shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects are found.

4.5 Product examination. Visually, dimensionally and manually examine each vulcanizers to determine conformance with the requirements of 3.2 through 3.4.4 and 3.6 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the vulcanizer and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.3), shall be cause for rejection.

4.5.1 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.2.1 and 6.4).

4.6 Tests.

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4.6.1 Performance test. Vulcanizers shall be operated in accordance with the manufacturer's instructions. The vulcanizer shall be connected to the specified source of power and allowed to be brought up to vulcanizing temperature in allotted time period. After vulcanizer reached vulcanizing temperature, the vulcanizer shall be used to apply a three inch diameter tube patch and a valve stem repair on an inner tube. The vulcanizer automatic temperature control shall be observed during this operation. Repairs shall be examined for imperfections, tubes shall be examined for damage, and functional ~~operational requirements shall be observed to determine compliance with the~~ following:

(a) Patch and valve stem repair performed properly (see 3.5.1).
(b) Functional requirements (see 3.5.1).
(c) Vulcanizing temperature time (see 3.5.1).
(d) Clamping unit operation (see 3.6.1).
(e) Automatic temperature control requirements (see 3.6.3.2).
(f) Specified source of power (see 3.6.3.1).

4.6.2 Temperature control and accuracy test. The vulcanizer shall be connected to a specified source of power and allowed to reach vulcanizing temperature in the allotted time period (see 3.5.1). Temperature of the plates shall then be monitored using instrumentation which has an accuracy of 0.5 degrees F. The temperature shall be varied from +287 degrees to +212 degrees Fahrenheit (F). The automatic temperature control shall maintain any temperature within this range to ±5 degrees F for a minimum 10 minute period.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Levels of preservation/packaging and packing.

5.1.1 Military. When specified, requirements for level A preservation/packaging and levels A and/or B packing shall be in accordance with MIL-V-45554 (see 6.2). When specified, requirements for level C preservation/packaging and packing shall be in accordance with MIL-STD-1190 (see 6.2).

5.1.2 Commercial. When specified, preservation/packaging and packing shall be in accordance with ASTM D3951 (see 6.2).

5.1.3 Civil agencies. When specified, unit packaging shall provide adequate protection against damage during multiple shipment, handling and storage. Packaged units shall be packed in a manner to afford adequate protection against damage during shipment from the supply source to the first

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receiving activity. Shipping containers shall comply with the rules and regulations applicable to the mode of transportation. The supplier may use his standard practice when it meets these requirements.

5.2 Marking.

5.2.1 Military and commercial marking. Marking of the following levels, military A, B or C and commercial shall be in accordance with the requirements of ~~MIL-STD-129 (see 6.2)~~.

5.2.2 Civil agencies marking. Requirements for marking for shipment for civil agencies shall be in accordance with the requirements of FED-STD-123 (see 6.2).

6. NOTES

6.1 Intended use. The vulcanizer is electrically heated and is intended for use in base and field shops to apply tube patches and valve stem repairs to inner tubes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Painting and finishing if different (see 3.8).
(d) Product identification if different (see 3.9).
(e) First article inspection if different (see 4.3).
(f) Quality conformance inspection if different (see 4.4).
(g) Level of preservation and packing (see 5.1.1).
(h) Type of marking required (see 5.2.1).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 OSHA standards. Products which meet all requirements of this specification comply with the OSHA standards published prior to the date of this specification and identified in Section 3 (see 3.2.2.1).

6.5 Definitions.

6.5.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, solvents, chemicals, and heat (see 3.4.4).

MILITARY INTERESTS: CIVIL AGENCY COORDINATING ACTIVITY:

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Custodians
Army - AL
Navy - YD

Review Activities
Army - ME, AT
DLA - CS

User Activities
Navy - MC

GSA-FSS

PREPARING ACTIVITY:

Army - AL

DoD Project 4910-0750

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the electrically heated vulcanizers shall be protected from the heat by being

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both t .e document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
Cmdr, USA Armament Res, Dev. & Engr Ctr		
c. ADDRESS (Include Zip Code) ATTN: SMCAR-EST-S Rock Island, IL 61299-7300	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) DSN
Cmdr USA Armament Res, Dev. & Engr Ctr	(319) 782-8230	793-8230
c. ADDRESS (Include Zip Code) ATTN: SMCAR-EST-S Rock Island, IL 61299-7300	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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